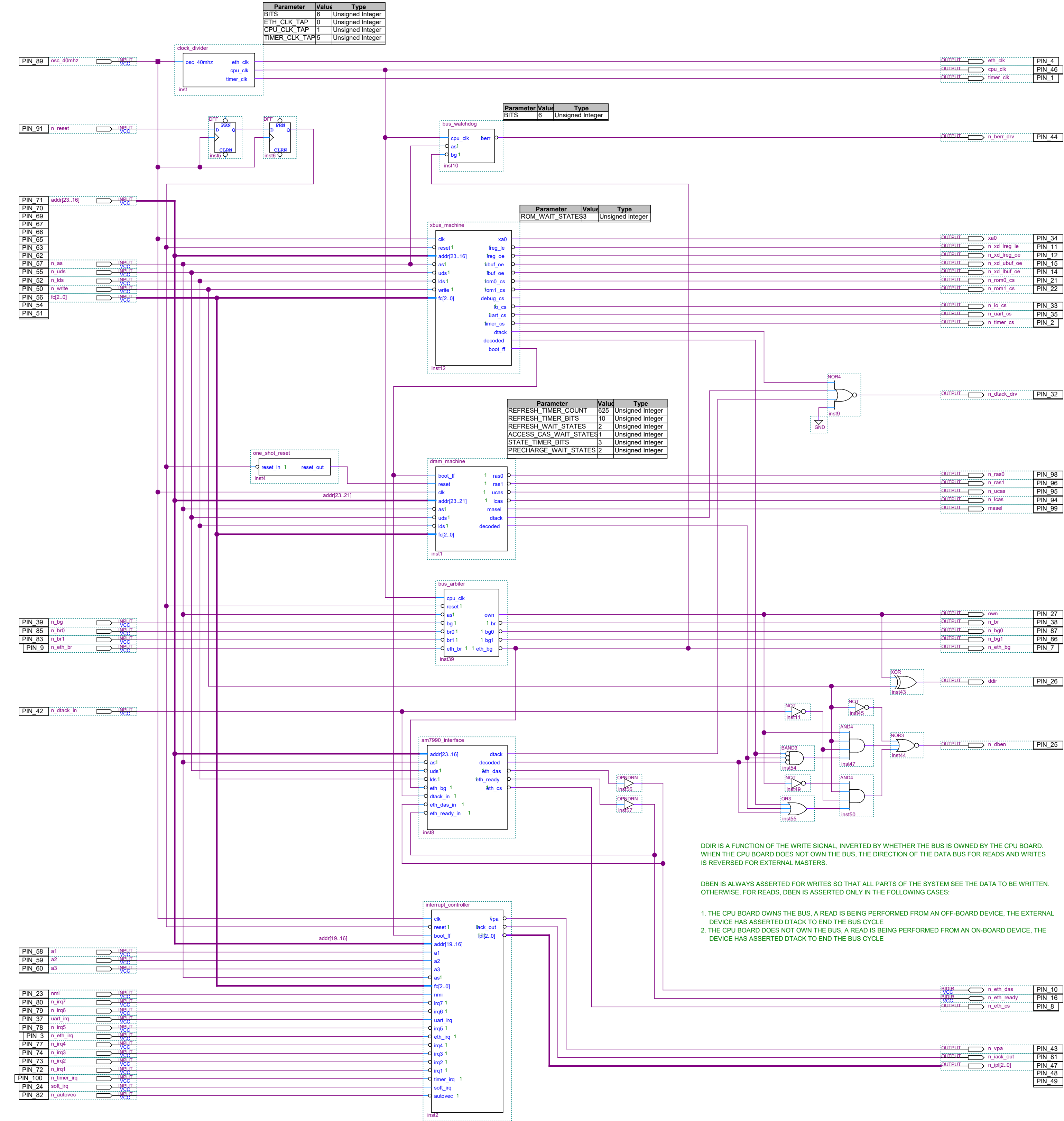




DDIR IS A FUNCTION OF THE WRITE SIGNAL, INVERTED BY WHETHER THE BUS IS OWNED BY THE CPU BOARD. WHEN THE CPU BOARD DOES NOT OWN THE BUS, THE DIRECTION OF THE DATA BUS FOR READS AND WRITES IS REVERSED FOR EXTERNAL MASTERS.

DBEN IS ALWAYS ASSERTED FOR WRITES SO THAT ALL PARTS OF THE SYSTEM SEE THE DATA TO BE WRITTEN. OTHERWISE, FOR READS, DBEN IS ASSERTED ONLY IN THE FOLLOWING CASES:

1. THE CPU BOARD OWNS THE BUS, A READ IS BEING PERFORMED FROM AN OFF-BOARD DEVICE, THE EXTERNAL DEVICE HAS ASSERTED DTACK TO END THE BUS CYCLE
2. THE CPU BOARD DOES NOT OWN THE BUS, A READ IS BEING PERFORMED FROM AN ON-BOARD DEVICE, THE DEVICE HAS ASSERTED DTACK TO END THE BUS CYCLE